***Student's Name:*** .....

**THE PRESIDENT'S OFFICE**

**REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT**

**ARUSHA REGION TERMINAL EXAMINATIONS - MAY, 2025**

**FORM THREE**

**SUBJECT CODE: 041** **BASIC MATHEMATICS** **Time: 3:00 Hours**

**INSTRUCTIONS:**

- This paper consists of sections A and B with 14 (fourteen) questions
- Answer all questions in both sections
- All necessary working and answers for each question must be shown clearly
- Mathematical tables and non programmable calculator may be used.

## SECTION A: (60 MARKS)

1. a) Write 784.3678 correct to
   i) Hundredths (ii) Hundreds (iii) Four (4) significant figures
   b) One third of the sum of two numbers is 50 and one fifth of their difference is 2. Find the numbers

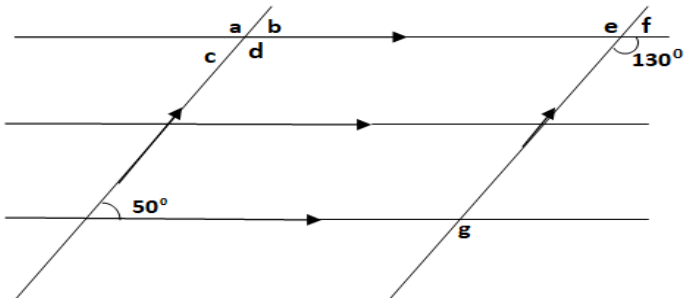
2. a) Find the value of x if $\left(\frac{2}{3}\right)^{2x-3} = \left(\frac{243}{32}\right)$
   b) Given that log x = $\overline{5}$.3567
   Find log $\sqrt[3]{x}$

3. a) In a class of 95 students, 60 students study mathematics only and 32 study physics but not mathematics and one students study neither mathematics nor physics. Find (i) How many students study mathematics (ii) How many students study both subjects, mathematics and physics
   b) Given that $\cap(A \cap B) = 10 + x$, $\cap(A) = 5x$ *and* $\cap(B) = 2x$ *and* $\cap(A \cup B) = 50$. Find the value of x.

4. a) The x and y- intercepts are 2 and 4 respectively. find the equation of the line in the form of ax + by + c = 0
   b) A translation takes point (-2,1) to (3, -5). Find where it takes (2,7)

5. a) The interior angle of a regular polygon is 120° greater than an exterior angle. Find the number of sides of a polygon.

b) Consider the following figure




Without doing calculations, Find the unknown angles marked a,b,c,d,e,f, and g

6. a) Mr. Ogango from Kenya fight against Covid – 19. The government of country M offered Us $ 1200 monthly to each family. If Mr. Mack's family, received the money for three months consecutively. Find the total amont

   i) Tanzania shillings $\begin{pmatrix} 1\ U\$ = 2280\ Tsh \\ 1\ Ksh = 22\ Tsh \end{pmatrix}$

   ii) In Kenyan shillings

   b) The resistance R of a wire is directly proportional to its length **L** cm and inversely proportional to its cross section area (A) $m^2$. When L = 2cm and A = 4 $m^2$, the resistance R is 8750 Ohms. Calculate the value of R when L = 36 and A = 3 $m^2$

7. a) The price of a radio set which includes VAT is Tshs 133,800. If the rate of VAT is 30%. Find the price of the radio before VAT was added

   b) Given that A:B = 3:4 and B:C = 5:7 Find (i) A:C (ii) A:B:C

8. a) If 4, x, y and 22 are in A.P, Find

   (i) The value of x and y

   (ii) The sum of the first $16^{th}$ term of an A.P

   b) The sum of third and fourth term of G.P is12. The sum of fourth and fifth term of G.P is 36.

   (i) Find the common ratio (ii) Find the first term

9. a) An observer on the top of a cliff 25m above the see level, views a boat on the sea at an angle of depression 60°. How far is the boat from the top of the cliff.

   b) Find the value of $\frac{Sin\ 60°}{\cos 60 \sin 30}$ without using the mathematical tables.

   c) If x is acute angle in a right triangle and that sin x = 0.6. Find the value of

   $$\frac{\sin x + \cos x}{\tan x}$$

10. a) Solve $ax^2 + bx + c = 0$ by completing the square method

    b) Solve $x^2 - 3x + 2 = 0$ by quadratic formula.

    c) Given that $\mathbf{a} - {}^{\mathbf{1}}/_{\mathbf{a}} = 10$ . Find the value of $a^2 + \frac{1}{a^2}$

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**SECTION B : (40 MARKS)**

Answer **ALL** questions in this section

11. The following frequency distribution table shows the monthly wages of 40 workers in a certain company

| **Wages (Tsh)** | **Number of workers** |
|---|---|
| 20000 – 29000 | 2 |
| 30000 – 39000 | 5 |
| 40000 – 49000 | 7 |
| 50000 – 59000 | 11 |
| 60000 – 69000 | 9 |
| 70000 – 79000 | 3 |
| 80000 - 89000 | 3 |

By using the mid-point of the class interval 50000 – 59000 as the assumed mean calculate the

(a) Mean wage (b) Mode (c) Median

12. a) Draw the graph of the inverse of the relation R= $\{(x,y): x+y \leq 1 \; and \; y \leq x-1\}$ and then find the domain and range of the inverse relation.

b) (i) Find the domain and range of the relation R = $\{(x,y): y = 2x^2 - 5x + 3\}$

(ii) Given that $x \in A$, where A = {0,1,2,3,4} draw a pictorial representation of the following relation R = $\{(x,y) : y = -x^2\}$

13. a) By using a sketch, find the image of point A(2,3) after a reflection in the line y = x followed by another reflection in the line y = -x

b) The vertices of triangle $\Delta \; ABC \; are \; A(2,3), \; B(1,4) and \; C(2,5)$ are reflected in y – axis

Find (i) The coordinates of the image

(ii) Sketch both $\Delta ABC$ and its image ΔA'B'C'

14. a) (i) Write the function f(x) = $2x^2$ + 3x + 1 in the form of f(x) = a(x + b)$^2$ + c where a, b and c are real numbers and a ≠ 0. Hence, find the turning point

ii) Given the linear function F(x) = ax + b. If f(2) = 6 and f(3) = 4. Find the value of a and b and then write down linear function

b) The function is defined as f(x) = $\begin{cases} 4 \; if \; -2 \leq x < 2 \\ 3 \; if \; 2 \leq x < 4 \\ 2 \; if \; 4 \leq x < 6 \\ 1 \; if \; 6 \leq x < 8 \end{cases}$

i) Sketch the graph

ii) Determine domain and range